Robot Dataset Audit Report

Dataset: MaxT90/so101-pick-place (public on Hugging Face)
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Date audited: Oct 11, 2026
Report type: Sample audit (portfolio)

Overall verdict: PASS after one fix: 49 of 50 tries usable (98%). No critical issues. One major issue (try 0, dead start) is fixable by trimming, with no re-recording needed. Two minor issues.

What this audit can't catch: wrong labels and frozen-but-decodable camera pictures. These are covered only by sampled human video review (2 of 50 tries watched in this sample: tries 0 and 45), not by the automated checks. Labels were not fully reviewed in this sample.

1. Dataset overview

Item	Value	Item	Value
Dataset	MaxT90/so101-pick-place	Frames	22,771
Link	huggingface.co/datasets/MaxT90/so101-pick-place	FPS	30
Robot	so_follower (SO-101 arm)	Cameras	2: wrist, overhead
Task	Pick and place	Tool	audit_checks_v1 (9 automated checks, incl. full video decode)
Tries (episodes)	50	Joints	shoulder_pan, shoulder_lift, elbow_flex, wrist_flex, wrist_roll, gripper

2. Checks run

Automated (9 checks, audit_checks_v1):

#	Check	What it asks	Result
#1-2	Episodes	Are all tries present, in order, none empty?	PASS: all 50 present (0-49), shortest 346 frames
#1c	Duplicates	Is any try an exact copy of another?	PASS: 50 unique
#3	Lengths	Is any try under half or over double the normal length?	**FAIL**: try 0 is 948 frames vs normal 438 (14.6 s); limits 219-876
#4	Timing	Steady 30 fps, no skipped frames?	PASS: steady 30 fps, 0 skipped
#5	Sync	Are the cameras in sync with the robot data?	PASS: 0.000 s max gap
#5b	Videos	Does every video decode end to end?	PASS: 22,771/22,771 frames on both cameras
#6	Action/state	Does the robot follow its commands?	PASS: mean gap 0.9-3.1; gaps >20 in at most 0.6% of rows
#7	Actions	No dead joints; clipping under 1%?	PASS: no dead joints, clipping under 1%; shoulder_lift, wrist_flex, wrist_roll go past ±100
#7	Empty tries	Does every try actually move?	PASS: least-moving try (24) is 88% of typical

Human review:

#	Check	How	Result
#9	Endings	Watched the try flagged by the end-pose hint	**REVIEW**: try 45, gripper left open (task finished)
#3 follow-up	Long try	Watched try 0 in the viewer, confirmed with per-second data	Dead start confirmed
—	Calibration	Calibration Presence Checklist v1 (5 yes/no), checked by hand	1/5 (only camera details present)
#8, #10	Camera clear, label correct	Sampled human video review	Not fully reviewed in this sample
—	Coverage	Coverage Rubric v1	Not scored in this sample

3. Findings

Levels: Critical: the robot could learn something dangerous or wrong. **Major**: that try needs fixing. **Minor**: good to fix, but it won't break training.

Level	Finding	Tries	Evidence
Critical	None	—	—
Major	Dead start: robot completely still for 0-16 s, then does the task normally	0	#3 FAIL (948 frames = 31.6 s vs normal 14.6 s); watched in viewer; per-second movement 0.0 for seconds 0-15
Minor	No calibration file; joint scale can't be verified	All	No calibration data in meta/; shoulder_lift, wrist_flex, wrist_roll exceed ±100 (#7); checklist 1/5
Minor	Gripper left open at the end	45	#9 hint; gripper ends at 14.5 vs median end 0.8; commanded action about 14.8; watched in viewer

MAJOR: Dead start in try 0

- **What:** the robot is completely still for 0-16 s (per-second movement of 0.0 for seconds 0-15), then does the task normally in about 15 s.
- **Where:** try 0, 948 frames (31.6 s).
- **Why it matters:** a policy may learn to idle or freeze before acting.
- **Fix:** trim the first 16 s or exclude the try.

MINOR: No calibration file

- **What:** no calibration data in meta/, and joints exceed ±100 (shoulder_lift, wrist_flex, wrist_roll), so the scale can't be verified.
- **Where:** all tries.
- **Why it matters:** nobody can confirm what the numbers mean, so others can't safely reuse the data.
- **Fix:** upload the calibration file. Calibration checklist score: 1/5 (only camera details present).

MINOR: Gripper left open in try 45

- **What:** the gripper ends at 14.5 (0-100 scale) vs a median end of 0.8. The commanded action was also about 14.8, so the operator didn't close it; this is not a hardware fault. The task did finish and the brick is in the box. It's the only try ending that open (next highest is try 2 at 6.2).
- **Where:** try 45, last about 4 s (13.0 s try).
- **Why it matters:** it's small, but a policy may learn not to close at rest.
- **Fix:** keep it, or re-record with the gripper closed at the end.

4. Unusable episodes

Try	Reason	Fix
0	Dead start: robot still for the first 16 s	Trim the first 16 s, or exclude the try

49 of 50 usable (98%), passes the 85/15 rule (at least 85% usable, i.e. at most 7 bad tries out of 50). Try 0 is unusable until trimmed.

5. Recommendations

1. Trim the first 16 s of try 0, or exclude it.
2. Upload the calibration file with the dataset.
3. Close the gripper before stopping each recording.
4. Widen variety in future collection: more object positions, objects and lighting.

6. Coverage score

Not scored in this sample. Coverage (object position, objects, lighting, each scored 1-3, total out of 9) is part of the paid audit.